Energy and Speed Are Related

Chapter 3 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **How are energy and speed related?**

KEY WORDS

speed — Speed is a measurement of the _____ an object travels over an amount of time.

KEY IDEA

KEY IDEA **For the SAME object, faster motion means MORE energy of motion than slower motion.**

MODEL IT

We compare the SAME object moving faster and slower. A greater change is evidence of more energy of motion.

SAME OBJECT · FASTER → **GREATER CHANGE** → **MORE ENERGY OF MOTION**

DRAW / LABEL

Draw the same ball at a LOW ramp and a HIGH ramp. Show which rolls away faster.

LOW RAMP (SLOWER)	HIGH RAMP (FASTER)

Which one caused a greater change? How do you know? _____

Measured vs Inferred

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MEASURED VS INFERRED

We do NOT directly measure speed or energy. We measure the change and infer the relationship.

MEASURED (I USED A TOOL)	INFERRED (I REASONED IT)
ramp height	relative speed (faster / slower)
cup displacement (cm)	relative energy of motion (more / less)

FAIR TEST

A fair test changes ONE thing at a time and keeps everything else the same.

In Ramp It Up!, the ONE thing we changed was _____

What we kept the same (the ball, the cup, the surface): _____

RAMP IT UP! EVIDENCE

RAMP HEIGHT (BOOKS)	CUP DISPLACEMENT (CM)
1 book	
2 books	
3 books	

Claim + Evidence

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Name: _____ **Date:** _____

MAKE A CLAIM

Claim: For the SAME object, when it moves faster it has _____ energy of motion, because it caused a _____ change.

Evidence from Ramp It Up!: the _____ ramp made the cup move the farthest.

In my own words — the relationship: the same object moving faster has _____ energy of motion.

GRAPH IT

draw or sketch a bar graph of ramp height vs cup displacement

CHECK YOUR THINKING

Why do we compare the SAME object instead of a heavy ball and a light ball?

CONNECT *Chapter 1 → 2 → 3: energy causes change; moving objects have energy; faster (same object) = more energy of motion.*